



NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.C.A. (CBCS) Degree Examinations

January 2024

22MCA105 – SOFTWARE ENGINEERING AND TESTING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Explain the importance of Software Engineering along with its basic definition.	5	L*2	1	1
	b) With a neat diagram, explain the Incremental Model and justify how it is better compared to Waterfall Model.	10	L2	2	1
	c) Justify the importance of Software Validation.	5	L3	2	1
2.	a) With a neat diagram, explain Requirements Engineering Process.	10	L2	2	1
	b) Describe the general model of Design process with a neat diagram	10	L2	2	1
Unit – II					
	a) Explain how Agile Development is different from Plan Driven approach.	5	L2	2	1
	b) Describe why Extreme Programming is a widely used Agile method along with diagram highlighting Extreme Programming release cycle.	10	L3	2	1
	c) With suitable examples, explain the need for Pair Programming.	5	L3	2	1
4.	a) Explain Functional and Nonfunctional Requirements with suitable examples.	10	L2	2	1
	b) Explain the Requirements Elicitation and Analysis Process.	10	L2	2	1
Unit – III					
5.	a) With suitable examples and correct notations explain the usefulness of Sequence Diagrams.	10	L3	3	1
	b) Explain Event Driven modeling along with the state diagram of a microwave oven.	10	L3	3	1
6.	a) Explain Object-oriented design using UML with System Context for Weather Station.	10	L3	3	1
	b) Explain the high level architecture of the Weather Station and also highlight some of the object classes for the Weather Station.	10	L3	3	1
Unit – IV					
7.	a) Explain the six essentials of Software Testing.	10	L2	4	1
	b) Describe the process of verifying the functional and the internal design.	10	L2	4	1
8.	a) With a suitable example, explain the Equivalence Partitioning method.	10	L2	4	1
	b) Explain White box test methods along with examples.	10	L2	4	1
Unit – V					
9.	a) Describe Unit and Integration Testing.	10	L2	5	1
	b) Explain Usability Testing along with its characteristics and test process.	10	L2	5	1
10.	a) Explain Test Execution and Evaluation Tools.	10	L2	5	1
	b) Briefly explain the approaches to organizing the test function.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
